

Philosophical Fundamental Problems

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Philosophy does not, like other sciences, enjoy the advantage of being able to build upon immediately firm presuppositions, upon indisputable fundamental principles; for in the whole of philosophy there is not a single proposition that stands so firm that it cannot, when treated on its own, be assailed by criticism and dissolved in dialectic. For that reason philosophy cannot very well parcel out its proofs in the way that mathematics, for example, can. Although mathematical proofs must be presented in strict order if full, scientific insight is to be attained, the mathematician nevertheless does not expose himself to any essential misunderstanding if, for one purpose or another, he treats a particular problem in isolation. Those who are competent know how the presuppositions can be proved, and those whose prior knowledge does not suffice content themselves with the result of the demonstration, trusting that the foundation is firm. But upon such a trusting acquiescence the philosopher, it lies in the nature of the case, dare not reckon. In philosophy no proof can stand alone; the treatment of a separate part must borrow its light from the whole, and the comprehensive system-proof must, so to speak, be transparent in every single demonstration. But it is now well known from the history of philosophy that the systems succeed one another, so that the preceding ones always fall before those that follow. This is indeed to some extent the case in the other sciences as well, but with the—admittedly very significant—difference that certain discoveries, methods, and modes of proof either remain unassailed or at least, under a further-reaching

elaboration, yield points of support, recognizable points of departure for further progress. It is otherwise in philosophy. Here a new point of view, a new thought, a new idea cannot step out into the world as a physical or natural-historical discovery can. When a philosophical system, with its principle and its peculiar method, falls prey to critical dissolution and loses its standing, the whole is recast in such a way that, in the new casting, one can hardly recognize what the abiding element really is that the preceding system deposited. If one regards the matter exclusively from this side, one would be tempted to believe that philosophy was no true science. But the view is one-sided and therefore misleading; the misleading element lies in the fact that one takes one's models from other sciences and applies a standard that does not fit. Philosophy must necessarily have its ideal peculiarity; what essentially belongs to the invisible world of thoughts cannot possibly bear the same marks as that which can be demonstrated from piecemeal-positing conditions or exhibited in visible facts. Seen, on the contrary, from the standpoint of the development of thought and of a consciousness growing in clarity, it shows itself that philosophy, in steady historical continuation, labors at the solution of certain ever-recurring problems; that every thinker who, by placing one or more of these problems in a light advantageous to inquiry, has made a contribution to the solution, remains standing as an eminent figure in the history of philosophy; and that his contribution continues to have effect, even though his school has long since ceased to exist. For a closer understanding of this, and in order, if possible, to contribute to a somewhat clearer conception of philosophy as a science, attention is directed to the problems set forth below.

I. The Problem of Knowledge

Among the philosophers who in more recent times have treated the question of the conditions and limits of human knowledge, Imm. Kant indisputably occupies the first place—if not by the results to which his inquiries led him, then at any rate by the manner in which he has posed the problem. Earlier thinkers—whether, like Locke, they laid the emphasis on the senses, or, like Leibniz, on the understanding—simply presupposed that we know from experience what experience is, that experience is something that is understood of itself; but Kant's "*Kritik der reinen Vernunft*" shows that experience precisely is *not* something that is understood of itself, that in order

to discover which cognitions we can draw from the sources of experience, we must go back to the inviolable fundamental conditions of cognition and begin with the question of *the possibility of experience*.

It is evident that the presuppositions which the knower himself must bring with him in order to be able to make an experience cannot be borrowed from experience, even though it is only with the help of experience that they can come to consciousness. What, then, are these presuppositions? Without sensation it is impossible to apprehend the sensible, and without understanding impossible to determine what is sensed: experience is conditioned by sensation and thought; but by what is sensation conditioned, and how does it stand with the understanding? All sensation rests upon an intuiting. To sense an object in space is just as impossible without the intuition of space as it is impossible to apprehend a change in time without the intuition of time. The pure intuitions, *space and time*, are therefore a priori necessary presuppositions for all sensation and thereby for all experience. In order to procure a positive content, thought must attach itself to sensation; for the understanding is no intuiting faculty, and the old metaphysics' attempts to open up for itself a supernatural world and, by pure understanding, to make discoveries of supersensible things are fantastical. But that sensation cannot bring it about that any experience whatever is made without the assistance of the understanding is seen from this: that the one sense-impression cannot be distinguished from the other without a discriminating activity, and the one representation is not determined by the other without a judging activity. The judging understanding would not be able to exercise its activity if it did not itself contain *a priori* those root-concepts, categories, by which logical judgments must necessarily be conditioned. Which these categories are, and how many they are, can be seen from the division of judgments in formal logic. Thus the hypothetical judgment shows itself to be grounded in the categories of cause and effect, and the disjunctive in the category of reciprocity. The a priori categories of the understanding are therefore indispensable for all thought and thereby for all experience.

The pure sense-intuitions and the a priori concepts must necessarily be fundamentally different, indeed heterogeneous, since the senses are heterogeneous with the understanding; the question then arises how it is possible that forms so heterogeneous can be united in the cognizing consciousness. We are referred to the imagination, which, on the one side intellectual, on the other side sensible, by means of the intuition

of time, furnishes an a priori schema necessary for the possibility of experience. It is not enough, for example, to have assured oneself of the validity of the concept of cause as a concept alone; here a condition is required for the concept's application to the appearances in the sensible world, whereby the determinate order of time acquires decisive significance.

By pointing out the subjective fundamental conditions holding for all experience, Kant found himself able to overcome D. Hume's empirical skepticism. Hume rejected the concept of cause as a kind of habit-concept to which no experience corresponds. That things accompany and succeed one another we can experience, but that the one is cause of the other we cannot experience; what we by habit call cause is as unknown to us as what we call force. Kant grants Hume this much, in that he demonstrates the absurdity of wishing to derive the concept of cause from experience; but inasmuch as he at the same time proves that no experience whatever is possible without the concept of cause, he asserts victoriously and once and for all this concept's a-priori-empirical validity so emphatically that, according to him, experience and cause must be said to stand and fall with one another. In particular cases of experience it can often be doubtful what the cause is of an event that takes place. When A, B, and C are accompanied by D, it may perhaps be A, perhaps B, perhaps C, perhaps A and B, perhaps A and C, perhaps B and C, perhaps A, B, and C in conjunction, in which the cause of D is to be sought; for the decision of such doubtful cases one may employ induction, but induction precisely presupposes that there must be a cause.¹

Here, be it noted, there is no question of the relation between the cause and the thing itself, which falls outside our consciousness, but only of the necessity

¹"Zwar scheint es, als widerspreche dieses allen Bemerkungen, die man jederzeit über den Gang unseres Verstandesgebrauchs gemacht hat, nach welchen wir nur allererst durch die wahrgenommenen und verglichenen übereinstimmenden Folgen vieler Begebenheiten auf vorhergehende Erscheinungen, eine Regel zu entdecken, geleitet worden, der gemäss gewisse Begebenheiten auf gewisse Erscheinungen jederzeit folgen, und dadurch zuerst veranlasst worden, uns den Begriff von Ursache zu machen. Auf solchen Fuss würde dieser Begriff bloss empirisch sein und die Regel, die er verschafft: dass Alles, was geschieht, eine Ursache habe, würde eben so zufällig sein, als die Erfahrung selbst; seine Allgemeinheit und Nothwendigkeit wären alsdenn nur angedichtet und hätten keine wahre allgemeine Gültigkeit, weil sie nicht *a priori*, sondern nur auf Induction gegründet wären... Freilich ist die logische Klarheit dieser Vorstellung einer, die Reihe der Begebenheiten bestimmenden Regel, als eines Begriffs von Ursache, nur alsdenn möglich, wenn wir davon in der Erfahrung Gebrauch gemacht haben; aber eine Rücksicht auf dieselbe als Bedingung der synthetischen Einheit der Erscheinungen in der Zeit, war doch der Grund der Erfahrung selbst und ging also *a priori* vor ihr vorher." Kritik der rein. Vernunft. Leipzig 1838. S. 201.

of the concept of cause for our experience. An arbitrary play of sensations and representations is only of subjective significance and yields no experience. If a set of terms a, b, c, d, and so forth were incessantly to alter their order and mutual position without any conceivable ground whatever, we should indeed be compelled to apprehend these changes in time, but the temporal succession would give us no concept of cause and, as a matter of course, no experience either. If the image of a house, for example, changed every moment in such a way that now it turned its ridge downward, now upward, and, by the transposition of its parts, took on the shape now of a tower, now of a whale, then we should have to seek the ground of the changes in our own imagination and could in no way refer such an image to any object of experience. For the sake of experience there must be something that compels us to place the one before and the other after; it is, in other words, the necessarily determinate order of time that expresses the relation between cause and effect. The necessity is a priori, for it is a conceptual necessity; with this necessity the object is determined by experience, and experience by the object.²

From this one will understand that Kant, with inflexible rigor of consequence, must make a divorce between the object of experience (the phenomenon) and the thing in itself. There must be something that, independently of us, gives our empty forms of cognition—the sensible as well as the intellectual—a real content and determines our apprehension of the object; but that the object, as we apprehend it, cannot possibly be identical with the thing itself is a straightforward consequence of the fact that, in passing through our subjective media of cognition, it must necessarily take on a subjective tinge. Our apprehension of the thing is an alteration of the thing: we see objects through colored spectacles.

The sharp separation between the thing in itself and its appearance finds recognition in the exact sciences, particularly in the physiology of the sense-organs. Helmholtz has³ emphatically stressed that our images and representations are effects which the

²"Der Begriff, der eine Nothwendigkeit der synthetischen Einheit bei sich führt, kann nur ein reiner Verstandesbegriff sein, der nicht in der Wahrnehmung liegt, und das ist hier der Begriff des *Verhältnisses der Ursache und Wirkung*, wovon die erstere die letztere in der Zeit, als die Folge, und nicht als etwas, was bloß in der Einbildung vorhergehen (oder gar überall nicht wahrgenommen sein) könnte, bestimmt. Also ist nur dadurch, dass wir die Folge der Erscheinungen, mithin alle Veränderung dem Gesetze der Causalität unterwerfen, selbst Erfahrung d. i. empirisches Erkenntniss von denselben möglich; mithin sind sie selbst als Gegenstände der Erfahrung, nur nach eben dem Gesetze möglich." Anf. Skr. S. 196–97.

³Handbuch der physiologischen Optik. Leipzig 1867. S. 442–flgd.

intuited and represented objects produce upon our nervous system and thereby upon our consciousness. Here there is a reciprocal action between an outside and an inside, and the cause of the phenomenon is precisely this reciprocal action. To ask whether the representation I have of a table—its form, its solidity, its color, weight, and so forth—agrees with the real thing is the same as to ask whether a tone is red, yellow, or blue. Things are determined by their properties, and the properties by reciprocal action with our organs of consciousness and with other things. Cinnabar is red, be it noted, for those who can distinguish colors, but black for the red-blind. Although the latter, according to the law of reciprocal action, have the same right in their judgment as the former, we nevertheless say in general: cinnabar is red; likewise we often ascribe to a thing a property as though it belonged to the thing in itself, although this property holds only for the thing's reciprocal action with other things, as when we say: lead is soluble, and tacitly understand: in nitric acid, not in sulfuric acid. The result is that our representations of things have only practical significance. Our designations, our words and expressions, are only symbols; out of these symbols a language can be formed, expressive of our observations, exact investigations, and scientific cognitions, but which has nothing, absolutely nothing, to do with the essence of things.

If this now is the final, decisive result of the inquiry, if the problem of knowledge is herewith once and for all solved—in principle, that is—if all further-reaching research will serve only for a further confirmation of the Kantian theory: then not only must the works of Fichte, Schelling, and Hegel be regarded as misguided, but every future philosophical attempt to illuminate the essence and ground of existence must be dismissed as barren and, for science, misleading. However the matter may stand, this much is certain, that the Kantian doctrine of knowledge has an ambiguity that must be drawn forth into the light. The fundamental question is: how can that which fills our forms of cognition with *real content be so absolutely heterogeneous with these forms that the thing itself must remain entirely unknown to us?*

Let us call the unknown X and confine ourselves to a definite object, for example a table. That we apprehend the table in space and ascribe to it a bounded extension, whereby it acquires its shape, is evident; but the pure intuition of space gives us no real table. The table is not a product of our own imagination; here is a something that acts upon our senses, notably upon sight and touch—this something is X. The table, the real phenomenon, thus comes about through two heterogeneous contributions,

a subjective one from ourselves and an objective one from the thing in itself. We can symbolize it by an equation: $X + Y = P_{x,y}$, where X denotes the objective, Y the subjective, and $P_{x,y}$ a function of X and Y , the phenomenon. If we set $Y = 0$, then the subjective vanishes, and $X = P_x$, that in the phenomenon which is due to the objective alone, remains. That P_x cannot itself be a phenomenon, that subjectivity is required in order to make the thing into a phenomenon, Kant has excellently shown; but how this happens, or even the possibility that it can happen, he leaves standing in complete ambiguity. — How, then, does X actually stand in relation to $P_{x,y}$, the space filled by the table? One can conceive the matter in several ways. X may be assumed, without filling any space, to make an impression on the soul of such a character that the determinate space-filling corresponds exactly to the intensity of the impression. The real table is thus by no means a space filled by X , for the X -impression brings it about only that subjectivity, which itself fills the given space, must imagine that it is a real object that fills it. This assumption must strike us as unreasonable, especially when we consider that the table is not merely an object of sight, but also an object of touch. Everywhere that we touch the table's surface, we sense the reality. But the reality of every sensation derives, after all, from X -impressions; there must then be just as many different X -impressions as the table's surface has parts, and these impressions—whose difference here is a difference of place—must be arranged and distributed in exactly the same way as the parts of the surface, that is, arranged and distributed in space. Yet neither does this assumption satisfy, for it would follow that to the phenomenon of a square table there would correspond, if not indeed a square X , then at least X -impressions that fit exactly the square, to the phenomenon of a round table X -impressions that fit the round, and so forth. If one separates the impressions of X , which make the spatial determinations real, from X itself as that which is outside space, then the X existing outside space has, after all, nothing whatever to do with the real spatial phenomena; its significance for a filling of the intuition of space is reduced to nil.⁴

⁴If one assumes that X , in a certain negative way, reflects itself in its X -impressions that refer to the intuition of space, just as the X -impressions reflect themselves in the objects of experience, then one gets an X -phenomenon behind every experiential phenomenon; but since the X that stands behind the X -phenomenon must in turn reflect itself in an X -phenomenon that removes itself still further from the form of space, one gets X -phenomenon behind X -phenomenon ad infinitum before one reaches the last X , separated from space, the pure "Ding an sich".

Similar difficulties present themselves when we consider the phenomenal changes that give the intuition of time real content. The substance-phenomenon shows us what *abides* in time, the accident-phenomena what *alternates* in time, and yet the substance must be characterized by the accidents, and the accidents by the substance. Now it is asked: is it one and the same X that determines the reality both of the substance-phenomenon and of the accident-phenomena, or are there here two kinds of X? If we assume the former, then we lay into X the very same characteristic difference as into the phenomena. As the X of substance it *abides* in time, as the X of the accidents it *alternates* in time; this will then, in other words, mean that the thing in itself, the unknown, is able to fill our intuition of time with real content only on the inviolable condition that it, both from the side of substance and from that of the accidents, enters into time, abides in time, and changes in time. But this assumption stands in manifest conflict with the Kantian presupposition. If we prefer the latter and posit one X for substance, another for the accidents, then the multiplication of X will also be carried further. It is then not enough to have a general X for substance in general; we must have a particular X for each particular substance, a sulfur-X for sulfur-substance, a carbon-X for carbon-substance, an iron-X for iron-substance, and so on; as regards the accidents, it will be doubtful how far we can make do with one X for each group of accidents, or whether just as many X's are needed as there are accidents in each group. The Kantian doubling—phenomenon and thing in itself—comes, in this light, strongly to recall the Platonic: the table and tableness, the horse and the horse-itself. However one turns and twists it, the following question remains to be answered: Can X enter into changes or can it not? If it cannot, then it keeps itself quite rightly outside time, but then all phenomena of change become X-forsaken, unreal semblances. If X can take part in the changes and give them the stamp of experiential reality, then this reality-giving must also in one way or another itself enter into time-determinations and take part in time. To take up the X-impressions that determine the changes into the intuition of time, and yet to let X be outside the intuition of time, is an absurdity, for it is a contradiction.

The Kantian doctrine of knowledge impresses upon us that our concepts of the understanding have validity only in the determination of the objects of experience and find no application whatever to the thing in itself. But when the thing in itself falls outside all conceptual determinations, how then can it be determined? It is

supposed not to be determinable by sensation, since it is strictly held outside space and time; it must therefore be determined by thought. Since now the understanding cannot possibly procure for itself any positive content other than what it receives from sensation, the consequence must surely be that the thing in itself can be determined only negatively. One would now suppose that the only determination that might be given of X would be this, that it excludes all determinations. It would then be just as indeterminate that it is as that it is not, just as indeterminate that it makes an impression on our senses as that it makes no impression, and so on. But in this supposition we are greatly disappointed. X becomes expressly determined by an act of the understanding; it becomes determined by a judgment, not by a negative, but by a positive judgment: *the thing in itself is*. Being, a concept of the understanding, is here, contrary to the theory's own prescription, applied to X. Being denotes reality, something positive; whence does the understanding get this positive? Not from sensation, for sensation gives only what can be apprehended in space and time. Nor from anything other than sensation, for here there is nothing else that can give the understanding the positive. So X must, then, be known indirectly by its effects, that is, by the sense-impressions of which it is the cause. But this assumption too the theory has excluded, for cause is a concept of the understanding just as much as effect. By determining the thing in itself as an unknown cause lying outside understanding and sensation—a cause that is to be thought as objective and yet cannot be thought as objective, a cause that in its effects, for the sake of reality, must be sensed and yet cannot be sensed—Kant has, instead of showing the possibility of experience, on the contrary proved the *impossibility* of experience.

If there is no objective form at all that can belong to the thing in itself, then this unknown must necessarily dissolve into a hazy possibility, a ὕλη ἄμορφος, and lack all objective self-subsistence. It is then so far from the case that the in-itself indeterminate possibility can be said to determine the objective content of our subjective forms, that on the contrary it borrows its apparent forms of reality from us and only through this loan acquires the semblance of something objective. — Here, however, is a consideration that can be brought forward from the Kantian side. Doubtless we must ascribe objective forms to the thing in itself, insofar as we ascribe to it objective validity, but the point is that we do not know these objective forms at all. To be sure, they cannot be forms like ours; but who can prove that there are no other

forms of existence than those that underlie our sensation and thought? Not to speak of the characteristic differences of our five senses—sight, hearing, smell, taste, and touch—both among themselves and among different individuals, or of the possibility that cognizing beings may have both more and sharper and finer senses than we, there is surely no human being who can vouch that our a priori intuitions and categories correspond even approximately to the objective form-determinations expressed in the thing itself. Is not the objectivity of space and time problematic? Must we not, even as regards objects of experience, make a distinction between the thing's concept and our imperfect, often faulty concept of the thing? — Here again we have the ambiguity.

It scarcely occurs to any reasonable person to deny the imperfection of our intuition of space when it comes to a clear and distinct apprehension of definite, many-sidedly combined and thereby involved relations in space. To construct in space is not exactly an easy matter. Should anyone doubt this, he need only make even a merely preliminary acquaintance with analytic geometry; he will then soon learn how easily a curve or a mathematical body, whose equation is known to him under certain conditions, can become unrecognizable either by a displacement of the coordinates' point of intersection or by the whole coordinate system's being altered. The more combined the figure is, the more difficult are the special relations to see through: here, then, manifold doubts can arise about the objectivity of spatial relations in the particular. But it is one thing to concede the imperfection of the representations and sensations conditioned by the intuition of space, and another thing to deny objective validity to the a priori form-determinations of the intuition of space. If a cognizing being descended to us from higher regions and declared that it could distinguish millions of parts in the space we took to be filled by a single part, we should by no means contradict it; but if it moreover maintained that it could distinctly see the circle as a square and the square as a circle, should we then be convinced that its intuition of space was more objective than ours? If the stranger gave proofs that he could not merely sense the ether, but even count 733 trillion vibrations in one second, he would astonish us in the very highest degree; but if he then gave as the reason that he was able to apprehend the simultaneous as successive and the successive as simultaneous, should we then dare to conclude from this that his intuition of time came nearer to the objective than ours? Between space and not-space, time and not-time, no intermediate forms are possible. How many dimensions space may appear to the sensing being to have is not

here the question; the question is of dimension and succession. By withdrawing itself from all determinations of space and time, of dimension and succession, the whole “objective, unknown world” must precisely become known to us as a single intensive point.⁵

In confirmation that our concepts of reason (ideas) find just as little application to anything objective in itself as our concepts of the understanding, Kant sets up four so-called cosmological antinomies, namely two mathematical and two dynamical. In this connection we confine ourselves to the mathematical ones. First antinomy. *Thesis*. The world has a beginning in time, and is, as regards space, enclosed within limits. *Antithesis*. The world has no beginning, and no limits in space; it is, in respect both of time and of space, infinite. Second antinomy. *Thesis*. Every composite substance consists of simple (uncomposite) parts, and there exists nowhere anything but the simple and that which is composed of it. *Antithesis*. No composite thing in the world consists of uncomposite parts, and there exists in general nothing simple in it. — Since the proof both for thesis and for antithesis in each of the antinomies must be granted the same validity, this mutual conflict of the transcendental ideas is supposed irrefutably to prove that we can just as little reach the objective by means of our reason as by the use of our understanding. Let us test the proofs. Does the thesis in the first antinomy stand in insoluble conflict with the antithesis? We cannot see that it does. If the world is from eternity, then there is, after all, no contradiction whatever in the fact that up to a given point of time an infinite series of successive world-states may have elapsed. To be sure, an infinite series cannot by any successive synthesis be completed in a finite time; it requires an infinite time in order to be completed. But from whatever finite point of time one looks back toward the infinitely distant beginning, one looks, after all, into an infinite time. If there is nothing contradictory in assuming an infinite series of represented terms, then there is nothing contradictory either in assuming an infinite series of real terms, provided the reality, be it noted, is presupposed to be from eternity. Why can the world not have begun in time? Because, says the antithesis, a time must then have gone before, hence an empty time; but in an empty time nothing

⁵That the air-waves are heterogeneous with the tones and the ether-waves with the colors proves only that a cause in the objective can become unrecognizable by entering into reciprocal action with the subjective; but what is incomprehensible herein proves nothing either against the concept of cause or against the objective validity of time and space. Of the influence of our organization there will be speech later.

can arise, since in no part of such a time is there any condition for a distinguishing of existence from non-being. But such an empty time is more imaginary than any imaginary magnitude otherwise is. With an imaginary magnitude, mathematically understood, one can nevertheless reckon; but an entirely empty time, a time in which there is no difference between the past, the present, and the future, is no time, and with no time one cannot reckon. As surely as the real world really has a beginning, its beginning must necessarily fall in a real time, and the real time begin with the real world. Since in this respect it makes no difference whether we push the beginning's point of time back half an eternity or a whole one, there is, as regards the world's beginning, no conflict between thesis and antithesis. — In considering the limitation and non-limitation of space we come to the same result. The real world is limited, for the outwardly real must at every moment have definite limits and to that extent be a finite. But the world in its finitude and limitation is also in real becoming, and can thus at every moment be on the point of overstepping every finite limit. It is the real world that makes space objective and sets objective limits in space; cosmic space has no peculiar being outside the world. The antithetic is built on the impossibility of completing the successive synthesis of the terms in an infinite series, that is, on the impossibility of comprehending all parts of an infinite totality. But, we ask, will then a limited whole, which contains only 7 parts, be more real than a whole that contains 7 trillion parts, because the successive synthesis in the first case is easy for us, but in the last case impossible? Surely no one will maintain that. But if we may assume a totality with 7 trillion parts, a world-whole with 7 trillion firmaments, to be real, notwithstanding that the successive synthesis requires an enormously long time, how then can there be any contradiction in presupposing the reality of an infinite totality, because a successive synthesis (more correctly: coming-into-being) of its individual parts must have required an infinitely long time?

For the *Thesis* in the second antinomy it is adduced that nothing substantial could be given if no indivisible parts existed; the substantially composite must therefore consist of uncomposite parts. In support of the *Antithesis* it is adduced that no composition of substances is possible except in space. The substance existing in space must necessarily have substantial parts; but space has no substantial parts, it is a *totum*, no real *compositum*. The contradiction is now supposed to be: that composite substances cannot exist without simple substances, and that simple substances cannot exist in

space. Instead of consisting of simple substances, the composite substance comes to consist of composite substances *ad infinitum*, and that is absurd. — This contradiction, however, is more imaginary than real. If simple substances and space are incompatible magnitudes, then not only experience but also mathematics is impossible, for then intuition, and with it sensation, is absolutely incompatible with thought. “It is here,” says Kant, “not enough, for the pure concept of the understanding of the composite, to find the concept of the simple; one must find the intuition of the simple that corresponds to the intuition of the composite, and this is, according to the laws of sensibility, hence also in the case of objects of sense, entirely impossible.” — This impossible, however, is not so evident as it might seem at first glance. A mathematical body, a cube for example, is an object of intuition; how does it stand here with the simple and the composite? The given cube can be intuited as a single existent, a continuum bounded by 6 square faces; but it can also be regarded as a composite and resolved into elements. If it is denoted by an infinite summation of represented corporeality-elements $x^2 dx$, then the third part of the intuited cube is an integral, determined by finite limits, of these elements. The face can be resolved into face-elements $x dx$, the line into line-elements dx , and these again into elements of a higher order, and so forth. In relation to the body the corporeality-element is the simple, just as the face-element is the simple in relation to the face, and the line-element the simple in relation to the line. Thought and intuition are, as regards the mathematical body, agreed that the simple must be relative for the sake of the composite, and the composite relative for the sake of the simple, that therefore the fixing of an absolutely self-existent simple is not an idea of reason, but a fixed idea. Should not the same hold for the material body? It is supposed, according to Kant, to be an absurdity “besides the mathematical point, which is simple, but no part, only a limit in space, to think of physical points, which are indeed simple, but yet have the advantage, as parts of space, of filling it by their aggregation.” But the alleged absurdity will not become evident to us. Let us only put atoms and molecules instead of mathematical elements, and aggregations instead of infinite summations: the fundamental relation between the simple and the composite will nevertheless remain the same. That the finite discretion is a parceling-out, a breach of continuity, must be conceded. But even though the atoms are, in a real sense, ever so indivisible—indeed even if, what for other reasons is an absurdity, they are assumed to be absolute simple substances—the intuition of space can nevertheless have not the

least to object against this. The atom's substantial simplicity by no means stands in conflict with its mathematical divisibility. The extended-in-space simple substance has, to be sure, just as many substance-elements as the space it fills has space-elements, that is, it has infinitely many, for the discretion goes on *ad infinitum*. But since the infinite discretion, as is expressly seen from space, coincides with continuity, what can intuition object against the simple substance—a real continuum—filling space, the formal continuum? Since the parceling-out is presupposed, the finitude of aggregation steps in place of the infinitude of summation; but since the real distance between real parts precisely gives the form of space the stamp of reality, the aggregation must surely be an eloquent proof of the formal objectivity of space.

Kant is fully convinced of the insolubility of his antinomies, “denn für die Richtigkeit aller dieser Beweise verbürge ich mich⁶,” but the insolubility presupposed, what is thereby gained? “That we must not think the phenomena of the sense-world as things in themselves.” That must be conceded. But there follows yet something else from it, something that it is precarious to concede; for when it is expressly added that we must not think the fundamental principles for the connections of phenomena as valid for the things in themselves either, then the understanding is lost, then it grows dark before the eye of reason. If we could still stop at the preliminary result, that in all our cognizing we must hold to our experiences and leave undecided how the things themselves in the world unknown to us actually are constituted: that might pass. But the antinomies drive us to an extreme that must either turn into bottomless skepticism or awaken despair. We are to assume things in themselves which, although they condition our experiences, nevertheless exclude all conditions of experience, since neither space nor time, neither unity nor plurality, neither ground nor consequence, neither cause nor laws, neither whole nor parts, neither simplicity nor composition have the least significance for these in-themselves unsensible and unthinkable things. These are hard terms.

For the understanding of the conflict of the ideas of reason in the antinomies, Kant adduces the following: “Two mutually contradictory propositions cannot both be false, unless the concept that lies at the ground of them both is in itself contradictory; for example, the two propositions: a square circle is round, and a square circle is not round, are both false.” But when the human consciousness has only a choice between one of

⁶Prolegomena. Werk. 3ter Band. S. 264.

two: either to let the ideas of reason or the “things in themselves” be “square circles,” then it is at least excusable if it once more ventures to assert the right of reason by choosing the latter. This choice is characteristic of J. G. Fichte’s “Wissenschaftslehre”.

We can briefly denote the deviation from the Kantian standpoint by exchanging the symbol $X + Y = P_{x,y}$ for $XY = P$. That this must necessarily give the inquiry another direction and lead to an entirely different result is evident. When Y, the subjective, becomes nil, that is, when the cognizing subjectivity, the Ego, vanishes, the whole becomes nil: no indeterminable object, no self-existent X, remains. By establishing the reciprocal determination of subjective and objective, Ego and non-Ego, Fichte has introduced a great fundamental thought into the history of philosophy, a fundamental principle of such incisive significance that it can as little be excluded from the theory of knowledge as the law of gravity from physics. Although Ego and non-Ego mutually exclude one another, it is nevertheless self-evident that X, the non-Ego, can as little be ascribed any self-subsistence valid outside the Ego as it can merge together with the Ego itself. As real opposite to the Ego, the non-Ego must therefore be an element in the Ego’s essence, a barrier that is indispensable for the Ego’s self-development. Only by the thinking Ego’s becoming conscious of its opposite, its barrier, as an unconscious presupposition that has proceeded from its own essence, does it become conscious of itself, and overcome the barrier theoretically. The objects of experience, the outward phenomena, present themselves immediately and with a necessity over which the Ego has no command, and thereby take on a stamp of outward reality; but that the Ego not only recognizes the forms and categories in which the objects present themselves as its own forms, but also recognizes the necessity with which they present themselves as a necessity of the Ego’s own nature-ground, is, in the “Doctrine of Science,” the solution of the problem of knowledge.

The critical illusion in Kant consists in this, that the unknown thing constantly appears as something self-subsistent in itself, objective, positive—indeed as the only unconditionally positive—notwithstanding that it shows itself from all sides that this positive can be determined only negatively. The negative is positive, the positive is negative; this contradiction, against which the critic defends himself with all his might, crops up everywhere and vexes the Critique. This illusion Fichte has seen through with clear consciousness, in that he has recognized the significance of the negative and ascribed to X an essential value as non-Ego.

But despite this unmistakable advance, the Fichtean solution is just as unsatisfactory as the Kantian. What from the objective side is impossible for Kant, namely to point out how the in-itself indeterminable X can fill the forms of cognition with an unsurveyable manifold of real determinations, the same is from the subjective side impossible for Fichte. As non-Ego, X is far too uniform to yield any explanatory ground sufficient for the diversity of the phenomena; a stone is a non-Ego, a plant a non-Ego, an animal likewise: animals and plants have, in relation to the Ego, the same significance as minerals and stones. The subjective reflection of Ego and non-Ego consumes all objective relations between the phenomena among themselves. Kant does at least get a world of experience, even if on an ambiguous, self-dissolving foundation. Fichte, who removes the ambiguity of the foundation, does not get even so much; he gets only a subjectively necessary shadow-world of barriers and non-Ego, a world so empty that the nature-content vanishes, and the philosophical interest in the realities of experience is lost.

The Hegelian philosophy lets the problem of knowledge founder in the absolute knowing instead of being solved. The opposition of the cognizing subjectivity and the object of cognition, which is the bearer of the problem, dissolves into the dialectical unity of thinking and being that flows through the Logic and issues in the Idea. In the Idea, the Absolute, subjectivity is sublated, that is, it is transformed into a co-determination, a moment, even if a moment of totality; and the same holds of objectivity, though with a noticeable preponderance over the subjective. Instead of an intuiting, thinking self-being that has the concept for its content and the concept's reality for its object, we get in Hegel a concept that comprehends itself, an Idea which, after having given itself subjectivity in the form of reflection, again sublates this its subjectivity and translates itself into the form of immediacy as objectivity in existing objects. The concept, in absolute unity with its content *the Idea*, cannot possibly maintain itself in two opposite forms at once; when the Absolute is posited as subjective Idea in the form of subjectivity, that is, of reflection, the objective is reduced to a moment and has vanished; when the Absolute is posited as objective Idea in the form of objectivity, that is, of immediacy, subjectivity has vanished: it must then comprehend itself. But when the objective is to be able to comprehend itself just as well as the subjective, then the self-comprehending concept is incomprehensible. If the absolute Idea is to be the concluding unity of subjective and objective Idea, then it

must either assume a form which, in being neither subjective nor objective, raises it above the extremes, or it must be one of those two opposite forms in which it holds the extremes conjoined. The former is, from Hegel's standpoint, inadmissible; for Schelling's indifference of subjective and objective is only an abstract presupposition, a non-ground, which stands far below the Ideas, and an over-subjective or an over-objective has no warrant in "the absolute knowing." As regards the latter, we must then choose either the subjective or the objective as the original form of the absolute unity; but the Hegelian philosophy makes the choice impossible for us. If one compares the Logic with the philosophy of nature, it appears as though the nature-Idea, the Idea in the objective form of immediacy, were the derived, and the Logic's Idea, the Idea in the subjectively reflected form, the original. But by comparing the philosophy of nature with the philosophy of spirit we are convinced that the matter stands the reverse; for logic in logical reflection presupposes spirit. There can be no logical reflection where there is no spirit; but nature is the presupposition of spirit, just as immediacy is the presupposition of reflection. Nature is spirit's "An sich," and nature is the objective; it is therefore the objective concept that—in an incomprehensible way!—comprehends the subjective. That Hegel, by the immanent dialectic of the ideal form-determinations, takes a giant step forward, that by going further than Schelling he gets negativity developed by another measure and in a quite different depth than either Kant or Fichte, must be conceded; but the Absolute itself he does not get cleared up. Despite all the talk of the self-thinking thinking, the absolutely self-knowing spirit, it is impossible to learn what the knowing itself, the absolutely knowing, is—whether it is absolutely subjective or absolutely objective, whether it is a self or a self-less process or perhaps neither of the two. It is a self-comprehending concept that is to clear up everything and bring the last remnant of the unknown to vanish; but far from having eliminated X, the incomprehensibly self-comprehending has transformed the Absolute itself into an X, an impenetrable riddle that can fittingly be denoted by an enigmatic symbol: $XY = X_{x,y}$.

If the problem of knowledge is to be solved, the following fundamental conditions must inviolably be held fast:

- 1) *There must be an insoluble opposition of subjective and objective, in which subjectivity itself, with its a priori forms, is the overreaching unity.*

- 2) *Cognition must rest upon so thoroughgoing a reciprocal determination of the subjective and the objective that no object can either be or be thought to be without a corresponding objectivization.*
- 3) *That imperfection of objectivization which has its ground in the independence of the existing objects from organically conditioned subjects must be capable of being remedied by subjective equivalents of ontological objectivization.*

Against Fichte and Hegel, Kant is, as regards the first fundamental thought, unmistakably right when he holds fast the opposition between the cognizing subject and the object of cognition, just as he also—from the side of finitude—is right in presupposing an unknown, a thing in itself, that does not resolve into our cognition. But in the manner in which he determines the thing in itself he is grossly wrong; for it is not by excluding the a priori subjective form-determinations that the thing withdraws itself from knowability⁷, but by its apprehension being modified according to the subjective peculiarity of the senses and of the sensing beings. The unknown thing that is the object of sensation is a thing with sensible properties just as well as the experiential phenomenon that comes to our consciousness by way of sensation. In sense-systems of different order, A, B, C, and so on, the same unknown X with its properties must necessarily present itself in different phenomena. In the A-system the properties $x_1, x_2, x_3, \dots$ will present themselves as $a_1, a_2, a_3, \dots$, in the B-system as

⁷The more recent Critique has, however, gone briskly forward, in that it has extended the influence of “our organization” to categories such as unity and plurality, substance and accident, subject and predicate, and so on. We are, it is said, naturally compelled to apply such categories on account of the peculiar organization of our understanding; but it does not yet follow from this that they are in themselves immutable. Enigmatic Critique! That the understanding must not be conceived as a faculty by itself over against the other faculties of the soul, that its activity must be dependent on organic dispositions, can be seen from differing endowment; the understanding of the individual may be disposed to one kind of intellectual activity without being in the same degree disposed to another. Nor can it be denied that a one-sided direction of the understanding may mislead one into fixing fundamental concepts as exclusively a priori which yet are demonstrably not so; one may think of the Pythagoreans’ philosophical conception of the decadic system, for example. But such missteps are, after all, to be corrected by the understanding itself according to the understanding’s laws. Those critics, on the other hand, who give themselves the air of being able to conceive—not merely an understanding infinitely higher than the human, for that we can all do—no, an understanding that annuls the difference between unity and manifold, cause and effect, subject and predicate, must assuredly distinguish themselves by a quite peculiar organization. “Metaphysics” they might now easily abolish; but how does it go with the understanding? In order to conceive an understanding that is absolutely heterogeneous with the human understanding, the human being must, after all, take leave of his understanding.

$b_1, b_2, b_3 \dots$, and thus in the first case give the experiential object P_a , in the last case P_b . The cognizing beings have in the one system just as much right as those in the other. But when the cognizing beings must all concede that the thing itself falls outside their cognition, on what then does the truth of cognition rest? The answer to this question leads us to the second fundamental condition.

If the object in itself, with its properties, its concrete determinations, could really stand independently outside all objectivization, and if the objective thus standing were the really true, then the cognition of truth would be impossible. But by working through the relation between Ego and non-Ego, Fichte has strikingly proved that the objective, far from being independent, is dependent on and determined by subjectivity. The exaggeration with which he asserts the pure Ego's original self-limitation, and the energy with which it itself posits its objects, furnishes a most necessary counterweight against the vulgar assumption that objectivization does not really concern the objects, but that it becomes subjectivity's own affair whether it objectivizes truly or falsely. When we now concede to Fichte that every real object must be a necessary product of an objectivizing subjectivity, but cannot possibly concede to him that the objects really are in themselves as we objectivize them to ourselves, or that our human objectivization should be able to give the things existence, then we resort to the third fundamental condition, in order to discover how the imperfection of the finite objectivization is to be remedied.

The universally valid ground of cognition coincides with the ontological ground of objectivization. When someone, for example, propounds the hypothesis of the rotating primordial vapor out of which our planetary system is supposed to have developed, he certainly does not assume that that vapor was an object of experience for human sensation, any more than his opinion is that what was then "a thing in itself" has since, through our experiences, been transformed into vapor. Nor does he assume either that the primordial vapor existed in our subjective intuition of space and rotated in our subjective intuition of time, nor that it is we who with a priori necessity have prescribed to nature the laws according to which the whole development has taken place. No, he rather assumes that all of it—the space, the time, the vapor, the motion—was objective without any objectivization whatever. Since it is now proved that the last assumption is just as meaningless as the first, what follows from this? That here in silence there must have been slipped in a subjective equivalent of ontological

objectivization! Hypothetically this is expressed thus: if we had been present and been able to survey the whole, then the matter must necessarily have appeared as the description runs. Thus: every time our limited subjectivity goes beyond its own bounds of experience, it becomes, for the sake of the objective, compelled to presuppose an intuiting that is not our intuiting, a seeing that is not our seeing, a thinking that is not our thinking, an objectivization that is not our objectivization; it becomes, in short, compelled to presuppose an ontologically objectivizing subjectivity.

From the ontological fundamental law of objectivization, symbolized by $XY = O$, where O denotes the object, it follows: that every objective moment must necessarily presuppose a corresponding subjective one, since $Y = 0$ gives $O = 0$, and that conversely every subjective moment in a real objectivization must be determined by a corresponding objective one, since $X = 0$ gives $O = 0$. Subjective and objective are, in the ontological sense, correlative concepts, but not coordinate: the objectivizing subjectivity is, as the overreaching unity of the opposition, a *dignitate prius*.

From the fundamental law it follows further that the truth of cognition rests neither on the objective by itself nor on the subjective by itself, but on the reciprocal determination of both. Let there be thousands, indeed millions, of diversely constituted sense-systems in the universe, still the a priori forms of cognition are the subjective fundamental forms of universal reason, and universal reason is common to all cognizing beings. The cognizing beings that, with the same sense-system, come under the same order must see that the concrete objectivization conditioned by sensation has all the truth it can and should have, when the reciprocal action between the subjective and the objective is precisely determined for the order in question. The dissimilar sense-systems are just as little able to awaken doubt about the universal validity in the peculiarity of objectivization as the red-blind man, despite the truth of his objectivization, is able to awaken doubt about the difference of colors among those who have color-sense. The truth is that the reciprocal action between subjective and objective must be different for subjectivities of different order; but *the reciprocal action itself is the truth*. To assume an X standing outside the reciprocal action, unobjectivized, for the really true, is a dream, an imagining; outside the reciprocal action there is no truth, for outside the reciprocal action there is, ontologically understood, nothing at all.

II. The Problem of Reality

While Hegelianism has sought to transform the world of realities into a system of metaphysical concepts, Positivism has, for a change, thrown all metaphysical ballast overboard, in order to hold exclusively to realities and facts. It is, as is well known, not the realities that conform to our concepts, but conversely our concepts that must conform to the realities; the wisest thing philosophy can do is therefore to give up the a priori problems, abandon its old leaky vessel, go on board the steamer of physics as a passenger, and assume the title of Positivism. In justification of so important a step, the school's celebrated founder, Auguste Comte⁸, has pointed out three stages of development (*trois états*) which the human mind must necessarily pass through on its way to the cognition of truth, *the theological, the metaphysical, and the scientific*. At the theological stage one seeks the cause of the phenomena in a supernatural will and lets the arbitrariness of the supernatural will determine the course of things without regard to fixed laws. At the metaphysical stage the arbitrariness disappears and gives place to such supersensible, self-valid entities as the Absolute, Substance, the Cause, the Idea, and so on. This intermediate stage has been necessary as a transitional step from the theological to the scientific⁹; but in itself the metaphysical standpoint is barren, as can be seen particularly from the endless, resultless disputes into which it dissolves. The positive philosophy, which stands at the scientific stage, now characterizes itself by this, that it regards the phenomena as subject to unchangeable laws, seeks to reduce these laws to the smallest possible number of fundamental laws, and considers every attempt to penetrate either to the first causes or to the so-called final causes as utopian and meaningless¹⁰.

⁸*Cours de Philosophie Positive; Tome 1mier, Paris 1830. 1miere leçon.*

⁹*La théologie et la physique sont si profondément incompatibles, leurs conceptions ont un caractère si radicalement opposé, qu'avant de renoncer aux unes pour employer exclusivement les autres, l'intelligence humaine a dû se servir de conceptions intermédiaires, d'un caractère bâtard, propres, par cela même, à opérer graduellement la transition. Anfrt. Skr. S. 13.*

¹⁰*Nous voyons, que le caractère fondamental de la philosophie positive est de regarder tous les phénomènes comme assujétis à des lois naturelles invariables, dont la découverte précise et la réduction au moindre nombre possible sont le but de tous nos efforts, en considérant comme absolument inaccessible et vide de sens pour nous la recherche de ce qu'on appelle les causes, soit premières, soit finales. Anfrt. Skr. S. 14. Further, S. 16: La preuve manifeste de l'impossibilité d'obtenir de telles solutions, c'est que, toutes les fois qu'on a cherché à dire à ce sujet quelque chose de vraiment rationnel, les plus grands esprits n'ont pu que définir ces deux principes — l'attraction et la pesanteur — l'un par l'autre, en disant, pour l'attraction, qu'elle n'est autre chose qu'une pesanteur universelle, et ensuite, pour la pesanteur, qu'elle consiste simplement dans*

To exclude metaphysics is fully justified when by the word metaphysics one denotes a science that puts imagination in the place of experience in order to discover the essences and *qualitates occultæ* of which real things must absolutely consist, unconcerned whether it agrees with reality or not. But against so superficial a conception of the concept of metaphysics one must protest in the name of science. Hegel's "Wissenschaft der Logik" is essentially metaphysics. Whatever imperfections may be pointed out in this Logic, one should nevertheless not accuse the author of fetching his concepts from the world of imagination. The concepts are so inherent in the things that no existing thing can escape corresponding to its concept. That "cultivation and discipline" of consciousness¹¹ which Hegel demands on behalf of Logic he can with full right demand on behalf of all the sciences. — The relation between physics and metaphysics we may determine thus: physics forms its concepts according to the observed facts, metaphysics develops its concepts out of the conditions necessary for all being and existence. To deduce the concretions of the concepts of existence from pure thought is impossible; but to give an account of empirical phenomena without going back to the a priori concepts is also impossible. Gold, silver, iron are called partly different *things*, partly different *substances*; what use is it now to denote reality, the given fact, now by the word *thing*, now again by the word *substance*, when one is not in agreement with oneself as to wherein the designation substance differs from the designation thing? In science one must after all know what one is oneself saying. In order to be a *definite thing*, the phenomenon must absolutely be a *thing*; in order to be a *definite substance*, it must be a *substance*. On this point physics has the advantage of consulting metaphysics. In metaphysics, the objective logic, it is not only made evident that the thing must necessarily reflect itself in the properties, and the properties in the thing, the substance in the accidents and the accidents in the substance; but it is also made evident wherein the reflection consists, and how different it is in the case of the thing and in that of the substance.

From this standpoint it will be instructive to see two such strong thinkers as Comte and Hegel try their strength with one another. Comte maintains that there is

l'attraction terrestre.

¹¹"Das System der Logik ist das Reich der Schatten, die Welt der einfachen Wesenheiten, von aller sinnlichen Konkretion befreit. Das Studium dieser Wissenschaft, der Aufenthalt und die Arbeit in diesem Schattenreich ist die absolute Bildung und Zucht des Bewusstseyns." Hegel. Wissenschaft der Logik. Berlin 1833. Einleitung. S. 47.

no Absolute; and he is right insofar as he looks only at existence and the existing. In the world of facts the one term stands against the other, the one atom exists beside the other. Everything is parceling-out, everything is relation. Hegel maintains that the only thing that in truth is, is the Absolute; and he is right insofar as he looks at the original unity of being with universal thought. The relative is not merely relative in relation to another relative; but all relations and relative terms have their subsistence in an encompassing whole. If the encompassing whole is itself relative, it must necessarily have its subsistence in relation to the relative term. But in this thoroughgoing relation it then relates, through all relativities, to itself, and precisely thereby comes to subsist as the Absolute. Comte maintains that all that is logical is subjective; and he is right insofar as he thinks only of formal logic, for formal logic with its abstractions is undeniably subjective. Hegel maintains that the logical is essentially objective; and he is right insofar as he starts from the essential unity of thinking and being. The determination of thought to which no being corresponds is just as empty as the determination of being to which no thought corresponds.

Comte assumes that thing and reality can very well be determined in themselves and present themselves to us without any objectively determining concept; but here he is wrong. We abstract, as it is said, our concepts from the things; but to abstract concepts from things in which there are no concepts is just as impossible as to pluck hairs from the bald man. Hegel assumes that the objective concepts in their immediacy are themselves things, and that it must therefore be possible to deduce existing facts from logical universal concepts; but here he is wrong. How unhappily every attempt at a construing of real things *a priori* must turn out, his philosophy of nature has more than sufficiently demonstrated.

Comte teaches that reality is reality, but not concept; Hegel teaches that the concept as concept is reality: in the clearly expressed opposition between Positivism and absolute Idealism the *problem of reality* can properly first be said to arise.

The problem is: that thing and concept must be assumed both to be the same and yet not the same.

That thing and concept must, in an essential sense, be the same can easily be seen. Take whatever thing or existing object you will, and you will convince yourself that the sum of all determinations—be they material or ideal, sensible or unsensible—must, since the determinations characterize the object and make the thing what it is, necessarily

belong to the object's concept. In its sum of determinations the object subsists; with a change in the determinations the object changes; with the dissolution of the determinations the object disappears. When it nevertheless seems as though there lay something more in the object itself than in its concept, this derives simply from a confusion of the object's concept with our concept of the object; for our concepts of concrete objects are far from being exhaustive. That gold's color, malleability, specific gravity, form of crystallization, slight affinity for oxygen, and so on, are properties that characterize this metal and to that extent belong to the concept of gold, there is surely no one who will deny. When we must nevertheless admit that we have only an imperfect concept of gold itself, the reason for this is easy to explain. We are, after all, still far from knowing all of gold's properties, and even if we knew them from experience, we could still not boast of knowing gold's substance so long as we are not able to derive the properties from one another mutually or from the common unity in which they, as accidents, have their subsistence. To be yellow, to be malleable, to crystallize in octahedra, to have affinity for chlorine, cyanogen, and so on, are in reality such dissimilar determinations that no mortal is able either to point out their law-determined dependence on one another mutually or to discover the real unity from which they all with necessity proceed. And yet physics must concede that color, malleability, crystal-form are just as much functions of the gold-atoms' quality and arrangement as gold's unequal affinity for chlorine and oxygen; in other words: physics' conception of the laws is not limited to certain relations between thing and thing in outward generality; but every thing with its properties, every whole with its parts, every substance with its accidents, every body with its molecular structure expresses a system of law-determinations in which term and relation reflect one another. Even the smallest parts can, in the all-sided reciprocal action, just as little fall outside the reflection comprehending its moments as they can fall outside the laws. Since now all determinations, both of the particular terms and of the terms' mutual relations, enter as real moments into the determining reflection, and the totality of the reflected determinations is one with the concept, it follows from this that thing and concept must be the same.

And yet concept and thing are by no means the same; they are, on the contrary, completely opposed to one another. What a difference between the concept of gold and the existing gold! The existing is immediately a fact, an object of experience; but

the concept as concept tolerates no object of experience, no residues of immediacy, whether of sense-impressions or of images or of representations: everything must be dissolved, melted, purified in that fire of negativity which gives the concept its ideality's unsensibleness and logical distinctiveness. From this standpoint Comte undeniably has ground to make a divorce between concept and fact; for what holds of the fact does not hold of the concept. One can crush a stone, but not the stone's concept; one can boil a plant, but not the plant's concept; one can shoot a bird, but not the bird's concept; the concept can be thought, and one can think *of* the thing, *about* the thing, but the thing as thing cannot be thought. The concept, the universal, can be expressed in words, but the thing's individual determinateness must be pointed at as at a This and This.

The immediacy that distinguishes thing from concept cannot be logical; for even though the logical can point beyond itself, this always happens within the sphere of logic: Hegel's Logic is an eloquent example of this. Nor can the immediacy be illogical; if the object-determinations were illogical and stood in contradiction with the logical, the object would either be concept-less or stand in contradiction with its concept. Nor can the immediate in the object be something that is added or joined from outside to the logical, as when in certain logics a distinction is made between the object itself and the concept expressed in the object. What indeed should this something be? The material, matter? But if matter with all its determinations falls outside the concept, how then can the concept be expressed in the material object?

When now the sought immediacy can be neither logical nor illogical nor joined to the logical in an outward manner, then it must surely be the not-logical, which the logical needs, a necessary complement with which it is saturated, in that it disappears; this complement is the antilogical.

If we seek the logical root from which the antilogical springs, we find it in the category of power. Power in its immediacy is the expression of reality, and the expression of reality's immediacy is again the antilogical. The reason we regard the objects as realities and ascribe to them an immediate, objective self-subsistence is precisely the impression of the power with which they, each singly, come to meet us. The objects of experience appear in this light like great and small potentates, who maintain their place in proportion to the power they have; only that, for the sake of immediacy, we must add that, without all logical reflecting, they are what they

have, and have what they are. In apprehending facts, things, and realities from this standpoint, we have, in consequence of the logical, come out beyond the logical. The opposition of thing and concept is so striking that, when the thing is posited, the concept has disappeared, and when the concept is posited, the thing has disappeared.

If we now go back to the fundamental law of ontological objectivization, which was treated under the problem of knowledge, then it becomes possible for us to point out the ground both of Positivism's and of Hegelianism's one-sidedness. In that the Positivists immediately make straight for facts and real objects, they entirely forget to examine the relation between object and objectivization. The law says: as the object is, its objectivization must be; as the objectivization is, the objectivizing subjectivity must be. Since the adherents of the positive philosophy quite overlook this law, they cannot, with all their interest in the exact investigation of reality, of course get their eyes open either for objectivization in power or for the objective real-concepts by which the realities are *a priori* determined. They are zealous for the cognition of the unchangeable laws, without a suspicion that the unchangeableness of the laws has its ground precisely in the system of real-concepts. The best proof of this is the fact that Stuart Mill, whose inductive logic has set empirical induction in the clearest light, himself admits that the laws do not demonstrably belong to the essence of things, that one may very well imagine regions of the heavens where "events follow one another by pure chance without any fixed law." Just as little have the Hegelians, despite their absorption in the dialectic of the objective concepts, been attentive to the fundamental law of objectivization. They have categories such as "Becoming," "Appearing," "Mediation," "Passing-over into its opposite," and the like; but the category of objectivization is not to be found in the whole system. Had Hegel but once fixed his sharp, penetrating gaze on the problem of objectivization, then we should surely have got another explanation of the transition from the subjective to the objective than is the case in that part of his Logic which he calls the subjective¹². If it

¹²"Der Schluss ist *Vermittelung*, der vollständige Begriff in seinem *Gesetztseyn*. Seine Bewegung ist das Aufheben dieser Vermittelung, in welcher nichts an und für sich, sondern jedes nur vermittelt eines Andern ist. Das Resultat ist daher eine *Unmittelbarkeit*, die durch *Aufheben der Vermittelung* hervorgegangen, ein *Seyn*, das eben so sehr identisch mit der Vermittelung und der Begriff ist, der aus und in seinem Andersseyn sich selbst hergestellt hat. Dieses Seyn ist daher eine *Sache, die an und für sich ist*, — die Objektivität." Hegel. Die subjective Logik. Berlin 1834. S. 170–171. When the subjective movement through mediation has reached objectivity, and the immediate totality of being-determinations stands before the logical consciousness as a factually given, it shows itself what

is already from a logical standpoint evident that the object's concept cannot possibly objectivize itself, that self-objectivization in the objectivization of another is and must be subjectivity's ineffaceable characteristic mark, then it follows irrefutably from this that a subjectivity which in thinking, reflecting, and comprehending holds itself strictly within the boundary of the logical will never be able to carry it further than to a logical objectivization of logical objects.

In the real world the objects are not logical; they are real, immediate, antilogical: their self-subsistence consists in immediate power, their mutual relations are power-relations. What now lies in this? As the objects are, the objectivization must be: power-objects must be objectivized in power; further: as the objectivization is, the objectivizing subjectivity must be: only an absolutely powerful subjectivity is able to let realities proceed from their possibilities and bear a world of objects that really stands by power. In the objects power is immediate as power in another; in the objectivizing subjectivity power is reflected as self-power. Self-power, the subjective power, is ideal and logical; the object-power, the objective power, power in another, is real and antilogical. The opposition of ideal and real power is the strongest, deepest

kind of subjectivity it was that comprehendingly found the object. It is by no means the substance comprehended in the causal relation that has transformed itself into judging, inferring, mediating subjectivity and thereby produced the objective; it is quite simply a human reflecting subjectivity, which has assured itself that there must be concept in the world of objects, and now — without in the least seeing how this concept has become object, contenting itself with the assurance that the object is a concept — regards it from the outside, and seeks therein to find and recognize the formal determinations that the subjective logic has developed. Here there can be no question of ontological objectivization. — “Darum ist das Erklären der Bestimmung eines Objekts, und das zu diesem Behufe gemachte Fortgehen dieser Vorstellung nur ein *leeres Wort*, weil in dem andern Objekt, zu dem sie fortgeht, keine Selbstbestimmung liegt. Indem nun die Bestimmtheit *eines Objekts in einem andern liegt*, so ist keine bestimmte Verschiedenheit zwischen ihnen vorhanden; die Bestimmtheit ist nur *doppelt*, einmal an dem einen, dann an dem andern Objekt, ein schlechthin nur *Identisches*, und die Erklärung oder das Begreifen insofern *tautologisch*. Diese Tautologie ist das äusserliche, leere Hin- und Hergehen; da die Bestimmtheit von den dagegen gleichgültigen Objekten keine eigenthümliche Unterschiedenheit erhält, und deswegen nur identisch ist, ist nur *Eine* Bestimmtheit vorhanden; und dass sie doppelt sey, drückt eben diese Aeusserlichkeit und Nichtigkeit eines Unterschiedes aus. Aber zugleich sind die Objekte *selbstständig* gegen einander; sie bleiben sich darum in jener Identität schlechthin *äusserlich*.” Die subjektive Logik. S. 183–184. Consistently, the development of the concept under this presupposition ought to have gone the way of experience, to have penetrated the method of natural science and thereby secured for itself the concepts' expression in facts; but instead the illusion of the self-comprehending concept giving itself objectivity gained the upper hand to such a degree that one, after having intoxicated oneself in the dialectic of the immanent method, has now quite given up the dialectical — although dialectic is precisely a thoroughgoing critique of ideality — and is close to forgetting “die absolute Bildung und Zucht des Bewusstseyns,” which philosophy can never dispense with, and which for so essential a part is nevertheless owing to the great master.

opposition that can be thought. Here is a real dualism; but the dualism is grounded in original unity, for the absolute self-power has the opposition within itself and masters its opposition.

The age-old question of the origin of the world-matter thereby comes under a new light. Aristotle lets, as is well known, matter and form be equally eternal, both belonging to that cycle of movements in which possibility alternates with actuality, and actuality with possibility; matter is the possibility, form the actuality. But this explanation is not appreciably better than the more recent Materialists' assurance that matter and force are equally eternal. What indeed is it, to be eternal, if it is not to be the absolutely original? Absolutely original is, after all, only that which has the ground of its origin in itself. But how is a being to be able to have the ground of its origin in itself, if it does not at the same time have a ground of origin for something other than itself? Aristotle describes the first mover, which is not moved, the Godhead, as the being that does nothing other than eternally to think the thinking of thinking (αὐτὸν ἄρα νοεῖ ... καὶ ἔστιν ἡ νόησις νοήσεως νόησις)¹³ and which, in order not to stain its perfection with anything imperfect, must exclude the possibility of being-other. But to exclude the possibility of being-other is to exclude matter. That such a conception of the original cannot withstand a critical test is easy to point out. If the eternal being is to think itself, then this must necessarily happen by a movement of thought in which it holds itself fast by distinguishing itself from something other than itself; merely in order to think itself it needs an other. If, moreover, its thinking is to be one with an active, real being, then this active self-being must necessarily contain the possibility of being-other; but the possibility of being-other is precisely matter. An eternal self-being without matter-being, an eternal matter-being without self-being, are hollow abstractions. The opposition of intellectual and real shows itself here in its first ontological shade. One may now, with Spinoza, attempt to lay the all-reality into the being's universal unity, conceive the self-being as the infinite substance (*causa sui*) with opposite attributes, thought for the ideal and extension for the material, and reduce the finite things to modifications, limitations, negations of the infinite: it does not help. The abstraction sucks the blood of the substantial, so that there remains neither self-being nor other-being, neither ideality nor reality, that

¹³Translator's note: the Greek (Aristotle, *Metaphysics* XII.9) reads roughly "it thinks, then, itself... and its thinking is a thinking of thinking."

has any self-subsistence to show. The positivistic way out, to draw a line through metaphysics and hold to the realities of experience, is, as shown above, impassable; unless the decisive step is to be taken in all thoughtlessness. For as regards the way out of thoughtlessness, there holds of metaphysics what Hegel somewhere says of reason: the best way in which one gets rid of it is: not to get involved with it at all.

If one is to hold fast reality without giving up thought, then the ontological fundamental law holds, that the subjective, self-reflected ideal must have the original power in order that the objective, the immediately real, may have the derived. An equalization of the mighty opposition between self-power and power in another cannot take place by subtraction and addition. One does not increase self-power by drawing something from power in another; nor can one strengthen the real by weakening or diminishing the ideal. On the contrary! The greater the reality the objective has, the mightier must the ideality be with which it is posited and determined. The objective world of facts, of realities, is, in the scientific sense, the strongest expression of the ideal power with which the ontologically universal self-being masters and comprehends itself in all and all in itself. All scientific truth stands and falls with the fundamental principle that the finite subjectivity, which objectivizes in impotence, must everywhere have its ontological equivalent in an infinite one, which objectivizes in omnipotence.

The transition from ideal to real, from concept to thing, corresponding to objectivization, has no intermediate determinations and can have none. How indeed should these intermediate determinations be constituted? Logical they cannot be, for one may enrich a concept with as many logical determinations as one will, it still remains, as concept, within the logical. Antilogical they cannot be either, for the antilogical enters only when the transition has taken place. The transition must then be a power-movement, a conversion of power's ideality into real power, a leap by which nothing is leapt over, a *logical-antilogical outburst of power*. Such an outburst of power is originally one with the transition from *possibility to actuality*—understood, be it noted, in such a way that the possibility lies in the concept, hence in the logical—*potentia ab posse*¹⁴—the actuality in the antilogical reality, hence outside the logical. Let us see it in some examples. He who is to walk with a light in a powder-magazine must be careful: he has an explosion to fear. That the explosion is a transition from

¹⁴Translator's note: the print here reads "polentia af posse" (in roman), evidently a misprint for *potentia ab posse*, "potency from being-able-to-be."

possibility to actuality, an outburst of power, surely no one will deny. So long as the powder remains in actuality, in antilogical reality—saltpetre, charcoal, and sulfur in mechanical mixture—the explosion is only a possibility that lies hidden in the powder's actuality; but when the explosion becomes actual, the powder becomes unactual. The oxygen from the sulfate of potash unites with the carbon into carbonic acid, the nitrogen develops in gaseous form, the saltpetre's potassium forms with the sulfur potassium sulfide—in short: the actual powder disappears; but the powder's concept, the real-concept powder, remains behind in the elements as possibility. The *sudden* is here only a sensible indication of the *qualitative*. The possibility of hydrogen chloride, a mixture of chlorine and hydrogen, can, by being placed in sunlight, show a sudden turn into actuality. But the same transition can also take place slowly, when the mixture stands in daylight; the transition can then be regarded as a successive summation of manifold small, indeed infinitely small, outbursts. The outburst rests on the qualitative: either the hydrogen chloride's possibility or its actuality, in other words: either the ideal or the real, either possibility or actuality, either concept or reality; an intermediate there is not. To wish to condense the concepts into realities or sublime the realities into concepts is a speculative bungling, in equal degree contrary to experience and sound logic.

The whole real world rests upon an original outburst of power. If one rejects the unity-power, the all-comprehending self-power, then the system of real-concepts disappears and with it all the possibilities of reason. If one assumes the self-power, then there follows with necessity, as lies in its concept and comprehension in power, an outburst of antilogical realities. The characteristic mark of the outburst's antilogical reality is, seen from all sides, *otherness*. The elements of actuality have a double-sided stamp of otherness. That they are the other of self-power is seen from this, that they constitute the real manifold, the swarm of atoms, the parceling-out of immediate power through which the mighty unity confirms itself. That there really is reality in this otherness is seen from this, that there are objective relations, that the elements relate to one another as other by other to other ad infinitum. We see it in the relation between matter and force. Matter with its atoms is a striking expression of power in another. It appears, on a superficial view, as though the atoms' existence in space were only a collective coexistence with no other relation than that which lies in the spatial relations. Consistently the Positivists must, presumably, acquiesce in this assertion. By

empirical induction one is supposed to be able to convince oneself that the one atom-group follows upon the other—usually!—according to a definite law. By taking motion as a fact and appealing to the law, one gets free of force; that is, force is transformed into a mathematical symbol, for example into the second differential coefficient in accelerated motion. At this standpoint, however, Materialism cannot maintain itself. The active relations between atoms and atom-groups are too conspicuous for one to be able to escape the impression of a real attraction and repulsion. The active relations now acquire a gleam of reflection, and the reality of reflection makes itself known in force. Then it is said that force, for example the force of gravity, is a property of matter. But when force, the reflected-in-relation, is the active, then it must surely also be the really actual, and what then is matter? One does not rightly know whether it is matter that acts by virtue of its property, force, or whether force is perhaps something particular, a kind of reflected reality that adheres to matter, the immediate reality. The last assumption is not satisfactory; so one goes further, draws the immediate reality under reflection, makes matter itself into a phenomenon of forces, sublimates it into forces and force-points. But—*naturam furca expelles*—the antilogical immediacy asserts its right; one is nevertheless compelled to recognize a side of force that is *opposite* to force, and to grant matter, the immediate, self-less otherness, its inertia.

The unity-expression for matter and force is power in another, understood in such a way that matter, the immediate in power, yields the existing terms, and that force is reflected in the active relations. In matter the concept founders, the existing stands in the concept's place; but that the antilogical in existing matter is nevertheless logically determined by its real-concept can be expressly known from force. That two masses existing in space can attract one another at a distance is certainly incomprehensible when one throws the concept away; the representation of an attractive force adhering to matter gives no explanation. Only when one takes the real-concept along does the matter become intelligible. The existing masses are only one-sided expressions of power in another. One supposes that one has reality's complete expression in each of the masses by itself, and is astonished that two such realities, which, as it seems, need nothing but the space they each fill, can have anything to do with one another. That is an illusion. To existing terms' actuality there belong active relations; the attractive force, an expression of active relation, belongs inseparably in the real-concept in which the masses' reality is originally grounded.

Let us pause a moment to hear an objection. From the standpoint of physics, and then especially of Positivism, it must surely hold that attraction is a fact, and that the discovery of the law of attraction, independent of any analysis of metaphysical real-concepts, is owing solely to empirical observation, calculation, and induction. To give the matter an appearance as though that which depends exclusively on experience could also be constructed *a priori* is a barren shadow-boxing, a relapse into antiquated metaphysics. This objection would be completely justified if it did not have its ground in a misunderstanding. Here there is no question at all of deriving real facts from our own imperfect, impotent concepts; on the contrary! our task is to lead the facts of experience back to the power-concepts, without which they, in their antilogical reality, would stand before consciousness as blind facts. If it is in conflict with the scientific method that one, with the help of experience, strives to derive the laws of things from their concepts, how then will one defend the procedure of mechanical physics? It is, after all, evident that the laws for the impact of hard and elastic bodies cannot possibly be found by way of experience alone, since no body is either perfectly hard or perfectly elastic; they must necessarily be derived from the real-concepts: hardness and elasticity.

From the ideal side the real-concepts are reflected in the all-comprehending self-unity; from the real side they are inherent in the things. It is the latter that in the present connection especially concerns us. The actual contains possibility to another actual; the actual gases oxygen and hydrogen contain possibility to water. In order that the possibility may be actualized, conditions are required; the conditions have the actual's immediacy and yet correspond in reflection to the possible. Chemistry has examples in abundance. When the salt is decomposed, that is, goes back into its concept, its possibility, the acid and the base step forth in antilogical reality, and conversely; nothing but outbursts of power in another. The real-concepts have a peculiar affinity for the antilogical; this affinity is originally determined by the relation between power in another and self-power. The chemist may concede that the oxygen and the hydrogen in a certain way perish in the water, namely insofar as by the combination they give up their particular properties, so that a new body with new properties arises; but that this can be regarded as a return of the realities to their real-concepts he will hardly concede. He will, in consequence of the atomic theory, on the contrary reject the transformation and assert the difference of the composite

atoms from the simple atoms. The water-atoms are decomposable, for each water-atom consists of one oxygen and two hydrogen atoms; but the atoms of oxygen and hydrogen are, as simple atoms, indecomposable. The ontological reversal from reality into concept, from concept into reality, becomes by this mode of consideration highly suspect. Yet wrongly. It is finitude's, otherness's, real discretion on which the emphasis falls, but this real discretion is so far from conflicting with the real-concept's logical essence that it on the contrary confirms it: only by the discretion's antilogical elements entering as reflexes into mighty comprehension does the concept acquire the character of real-concept. The composite atom cannot possibly be regarded as a mere arithmetical sum of two or more simple atoms; for the sake of aggregation alone a mutual attraction must here be presupposed. Even if the attraction were—what, on account of the peculiarity of the affinities, it cannot be—merely mechanical, then the mechanical attraction, according to what we have seen in the foregoing, would alone be sufficient to assert the concept's validity. The oxygen and the hydrogen could not possibly combine chemically if they had no affinity for one another; but mutual affinity they could not possibly have if their real-concepts did not correspond to one another; but the real-concepts could not possibly correspond to one another if they were not originally comprehended in one and the same comprehension. In the oxygen, the hydrogen, and the water we have three real-concepts, two co-concepts and a concretion-concept determined thereby. When the co-concepts exist with antilogical reality, then the concretion-concept is a possibility; when the concretion-concept exists, then we have in this reality the co-concepts' possibility. The transition from possibility to actuality presupposes conditions, but takes place by an outburst of power.

It has been intimated above that the positivistic assurance of the immutability of the laws of nature stands, according to St. Mill's admission, on rather weak feet. But even if this weakness is overlooked, even if it is conceded that a merely empirical induction can give us a maximum of probability that a B will invariably follow upon an A, is any essential insight thereby gained? By no means. There is a difference between *certainty* and *insight*. Suppose the relation between the hypotenuse and the legs had been found practically by a measurement of the triangle's sides and squares constructed thereon, then repeated measurements of right-angled triangles could indeed give a certainty about equal to that which a geometrical proof gives; but only the proof gives insight, the measurement by itself gives no insight. All insight is conditioned by this, that the

representation of the one encloses the requirement of the other, that the one cannot be posited without the other having to follow, that the reality of the one bears within it the concept of the other. If the hypothetical proposition “if A occurs, then B must follow” is held fast merely empirically, then there is here no insight, however great the certainty may be. One will ground the phenomenon by enumerating its conditions, but however many conditions and intermediate determinations one enumerates, one still does not reach the ground; for it stands with the conditions as with the conditioned: the individual conditions are themselves phenomena. Instead of a grounding of the phenomenon one reaches only a summation of phenomena, which each in turn must be grounded by a summation of phenomena; the grounding is a continued demand for grounding, but the ground flees away into infinity: insight is never reached. One can, with a maximum of probability, get certainty about many things and about the connection of many things, without however attaining insight, rational insight into any thing.

A rational insight into the connection of things is only possible under the presupposition that there is reason in the things, that the things are realized concepts. The concept realized in the thing is the thing’s nature. That every thing acts according to its nature means that every thing acts in consequence of its concept; but what is the acting cause? For this question the problem of concept and reality presents a new side. The concept seems not to be able to be cause: the concept of gravity does not weigh, the concept of fire does not burn, the concept of a stone kills no dog. It must then be the thing that is cause. But when the thing can act only in consequence of its concept, then the concept must surely be the cause of the thing’s being a cause. What does this lead to? Quite directly to a distinction between two kinds of causes, ideal and real, and thereby to a more penetrating investigation of the relation between *ideal and real causality* on the whole. Such an investigation is of importance for the sake of insight and scientific cognition. To make matter and force into causes, without troubling oneself further either about the concept of matter or about the concept of force, is an easy matter; to make the real cause logically one with the concept of the cause is not exactly difficult either; but in the first case one excludes all insight, in the last all reality.

The logical-antilogical outburst of power, which transforms the concept into reality, transforms at the same time the ideal causality into the real.

That the cause can act only by virtue of the unity of concept and reality is seen from this, that acting and activity can take place only in consequence of the same unity. The thing as thing in antilogical immediacy cannot act. And why not? Because immediacy, sharply and exactly held fast by itself, means that the concept's content, and therewith possibility to another, is exhausted in the thing's existence; the only thing the thing in this case is able to do is: to exist. With exhausted concepts the things would exist side by side, excluding one another, but for the rest have not the least to do with one another. Herbart's absolutely positive reals can serve as an example of this. This acute thinker, persistent in exact analyses, suffered from the fixed idea that unconditionally independent reals should be able to be affected by one another without the positive in them betraying anything negative. A curious contradiction¹⁵! Under the mutual affecting the reals must precisely lose their absolutely positive character and sink down to becoming relativities. That it can no more be the concepts which, held fast by themselves in logical form, produce the things, is self-evident.

¹⁵The propensity to give empty verbal explanations instead of a real interpretation of facts, and the weaknesses of which metaphysicians, for the sake of the system alone, often make themselves guilty, F. A. Lange (*Geschichte des Materialismus und Kritik seiner Bedeutung in der Gegenwart*. Iserlohn. 1866. S. 463.) has aptly characterized: "Das absolut Einfache ist auch keiner inneren Veränderung fähig, weil wir uns diese nur in der Form wechselnder Ordnung der Theile denken können. Deshalb sagt Herbart auch nicht, dass die Realen auf einander wirken, sondern dass sie Einwirkungen von einander erleiden *würden*, wenn sie diesen nicht durch einen Akt der *Selbsterhaltung* Widerstand leisteten. Als ob damit etwas andres gesagt werden könnte, als mit der Annahme einer einfachen Wechselwirkung! Waitz legt in seiner Psychologie (S. 81) viel Werth auf den Unterschied zwischen *Dispositionen* zu einem Zustand und wirklichen Zuständen. So geht es in der Metaphysik. Zustände darf die Seele nicht haben; bei Leibe nicht, sonst ginge ja ihre absolute Einheit verloren! Aber Dispositionen, das ist ganz etwas andres; „Strebungen“, warum nicht? Der Metaphysiker widerlegt mit einem enormen Aufwand von Scharfsinn alle möglichen andern Ansichten, und wo er seine eigne Meinung entwickelt, schießt er einen logischen Purzelbaum von der gewöhnlichsten Sorte. Jeder Andre sieht, dass eine Disposition zu einem Zustand auch ein Zustand ist, dass Selbsterhaltung gegen eine *drohende* Einwirkung nicht ohne eine, wenn auch noch so feine *wirkliche* Einwirkung denkbar ist. Der Metaphysiker sieht dies nicht. Er hat sich mit seiner Dialektik an den Rand des Abgrundes getrieben, alle Begriffe hundertmal herumgewendet, hervorgezogen, weggeworfen, und endlich *muss durchaus und durchaus etwas gewusst werden*. Also die Augen zugedrückt und den *salto mortale* herzhafte gemacht — von den Höhen der schärfsten Kritik hinab in die allergewöhnlichste Verwechslung von Wort und Begriff! Ist dies gelungen, dann geht es munter weiter. Je mehr Widersprechendes in die erste Grundlegung aufgenommen wird, desto freier lässt sich schliessen, wie man denn bekanntlich aus mathematischen Sätzen, welche den Faktor Null in versteckter Weise enthalten, oft die merkwürdigsten Dingen ableiten kann." It is of course self-evident that a metaphysician's error can as little be charged to metaphysics as a mathematician's miscalculation to mathematics. If the mathematician subtly defends his errors, when they are clearly pointed out, instead of openly admitting them, he does not thereby compromise mathematics, he compromises only himself. The same holds of the metaphysician and metaphysics.

The concept of water can as little produce water as one can drown in the concept of water. The thing's concept is its possibility, a possibility that lies hidden in other things' actuality. Only by the actual things having possibilities in themselves to other things are they able to act. Of several freely existing masses each single one has in its actuality the possibility of changing the others' place in space; it is this unity of actuality and possibility that is expressed in the activity we call mechanical attraction. Two pieces of wood cannot be rubbed against one another without the possibility of the one having to correspond to the actuality of the other; light-activity cannot be given unless the light-giver's actuality corresponds to the ether's possibility, and conversely. Since now actuality is reality, and possibility concept, it is seen from this that ideal and real activity, ideal and real cause, must presuppose one another mutually.

The unity of ideal and real causality is a unity in opposition, an opposition-unity. From the side of reason the natural order of things makes the impression of an infinite, in all parts coherent whole; not even the smallest part can withdraw itself from the power of necessity, for everything subsists in infinite reciprocal action; it is a world of inner, essential relations, a world of concepts of ideal causality. From the standpoint of the sensible experiences, on the other hand, the world of realities shows itself as a world of chance, of outward conditions and real causality. According to the connection of concepts there has, for example, from a combination between silicon (silicium) and oxygen arisen silicic acid, from a combination between potassium and oxygen potash, from a combination between aluminium and oxygen alumina, and from the mutual combination of these combinations silicate of potash-alumina (feldspar)—all essentially in consequence of ideal causality. The one-sided concepts are contained in the more all-sided ones, and every trace of chance seems to have disappeared. By analysis of the concretion-concept feldspar we come to the co-concepts silicic acid, potash, and alumina; by analysis of these still concrete co-concepts to the elementary concepts of oxygen, silicon, potassium, aluminium—with reflected necessity on account of essential relations. If we begin with the elementary concepts and apply synthesis, we get the same necessity within the logical. But in the antilogical sphere of the realities something quite different shows itself. Here we meet circumstances, chances, and contingencies on account of outward relations. The oxygen and the silicon can here exist each by itself, without silicic acid arising; silicic acid and potash are found each

by itself, without silicate of potash forming, and so forth. In the circle of the outer conditions every outburst of power is dependent on intervening chance. The outburst, antilogical in the contingency, is in the necessity logical: the reciprocal determination of contingent and necessary characterizes the law.

This leads us a step further. There is in the system of real-concepts a difference between concept-values, marked by this, that some concepts apparently lie nearest to necessity, others nearest to contingency. Concepts such as earth, water, salt, acid, base, and the like appear as essence-concepts, substance-concepts, concepts of necessity; while others, such as: gravel, heap, aggregate, conglomerate, and so on, are formed in an entirely contingent manner by the coincidence of concept-consequences that do not essentially concern one another; such concepts are owing to contingency to such a degree that they do not even seem to deserve the name of concepts. This distinction between concepts of substantiality and of accidentality has a claim to attention, for the solution of the problem of reality is dependent on how it is interpreted.

If we begin from the contingent and concede that contingency makes the concept nominal, then the concept gravel is only a denomination for a heaping-together of parts, but no objective real-concept. The Positivist will smile when he hears that the concept gravel too is to be included in the eternal comprehension and have its ontological origin from an all-comprehending self-power. But when the concept gravel is nominal, what then is the concept granite? Also nominal! When the concept granite is nominal, what then is the concept feldspar? Naturally also nominal; the formation of feldspar has, after all, been conditioned by outward relations. That this mode of consideration is false shines forth from its consequence; for the consequence is that all objective real-concepts must disappear and give place to nominal designations of concept-less facts. The absurdity and superficiality herein has been sufficiently pointed out above; we are then compelled to begin from necessity. Just as in all real necessity there is an antilogical moment of contingency, since the really necessary is always *conditioned*, so there is in all the contingent a logical moment of necessity; if it were not so, there would open up in existence a wide field where the contingencies could carry on their play without regard to the laws.

There have been carried on, in the history of the sciences, particularly of philosophy, extensive disputes about whether, besides the physical causes, one could also assume final causes and ascribe to these objective validity. By distinguishing between

the outer and the inner purposiveness, Kant has held that, for the organic formations, one ought to make—not indeed a constitutive, but nevertheless a regulative—use of the principle of purposiveness; but even the regulative use seems more and more to sink in credit. The obscurity has its ground in the fact that the relation between ideal and real causality has not been sufficiently investigated.

When one without further ado separates the things from their concepts and, with denial of the ideal causality, imagines that one has enough in the physically acting causes, the final causes cannot possibly acquire any objective significance. But that this tearing-loose of the real from the ideal is unscientific will gradually become evident through a more thorough treatment of the problems of knowledge and of reality. Both the law of objectivization and the reciprocal determination of the logical and the antilogical in the outburst of power prove that the real causality has its ground in the ideal. When one one-sidedly clings to particulars, contents oneself with dissection, and blindly makes the whole into a result of the parts, then it is quite true that all talk of purpose and end must become nominal and superficial. Every finite attempt to fix something as end and something else as means founders on the infinite reciprocal action of conditions and conditioned terms; the alleged higher sinks down to the level of the supposedly lower, and conditioned necessity becomes sole ruler in the realm of things and laws. If reason is to come into its right, if a scientific insight is to be possible, then every approach to the solution of the great problems must at all points point toward a development of the parts out of the whole, toward the grounding of the real totality in the ideal, toward a world-design with an all-comprehending subjectivity as its author.

In the real-concept *design* lies the unity of ideal and real causality, in the concept of design purpose, and in the concept of purpose again the concept of means. The purpose, originally seen, is that requirement of subjectivity that the plan in the whole, the definite design, shall be actualized through a reciprocal development of the parts among themselves. The design itself is a system of designs: no part and no whole can give itself design, but every part, every whole acts and must act by virtue of its peculiar design. There must, in the name of reason, be a whole in order that there can be parts, there must be a higher in order that there can be a lower, there must be purpose in order that there can be means. The unscientific arises only when the finite subjectivity presumes to survey the plan, determine the aims, and apportion sun and

wind between objective purposes and means.

The fundamental question is how the subjective in reality relates to the objective. This problem itself comes into existence with the organic formations. What is a living organism, indeed what is life? Life is a conditioned self-activity, and the product of this activity is the organism; that is the shortest answer we can give here. The self-activity conditioned by matter exhibits a self-working-out, though restless, equilibrium of ideal and real causality. At the lower, elementary stage of development, the ideal causality, the ontologically subjective, is—what we have often enough emphasized—present everywhere, but so hidden in realities that the things act as matter-subjects by their properties upon one another: masses, molecules, atoms are real causes, physical otherness-causes on account of the hidden self-cause. With life's organizing activity the relation is reversed: the otherness-causes sink down to conditions and means for the self-cause coming to light, the self-actualizing ideal subjectivity, which in reciprocal action with the manifold of matter-subjects makes itself the ruling center and shows what the purpose is. In an enigmatic manner the double-sided causality separates into two relatively independent causalities, two heterogeneous principles, the ideal and the real; this must then lead the inquiry further.

III. The Problem of the Heterogeneous Principles